

Chapters 4 and 5, A Few Solutions

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4.7 (a) The linearized equations are

$$\frac{\partial h'}{\partial t} + H \nabla \cdot \mathbf{u} = Q, \quad (1)$$

$$\frac{\partial \mathbf{u}}{\partial t} + f \mathbf{k} \times \mathbf{u} = -g \nabla h' - \alpha \mathbf{u}, \quad (2)$$

where we've written $h = H + h'$. We'll drop the time derivative terms since we're told to assume steady state, but just write them above for completeness. I find it easiest to start from the linear mass equation, which in steady form immediately allows us to solve for the divergence:

$$\nabla \cdot \mathbf{u} = \frac{Q}{H}.$$

Since the forcing is axisymmetric (depends only on r , not θ), we can assume the solution is too. That means the radial velocity, u_r , is related to the divergence by

$$\frac{1}{r} \frac{\partial}{\partial r} (r u_r) = \nabla \cdot \mathbf{u},$$

which, integrating, gives the general solution

$$u_r = \frac{Qr}{2H} + \frac{C}{r}$$

where C is an integration constant. The value of C may be different in the two regions $r > L$, $r < L$. Requiring all velocities to be finite (there is dissipation in this problem, so no singularities) requires us to set $C = 0$ for $r < L$. For $r > L$, $Q = 0$, but u_r has to be continuous at $r = L$, or else there would be infinite divergence there, which would violate mass conservation. Thus we must require $C = QL^2/2H$, so the total solution for u_r is

$$u_r = \frac{Qr}{2H}, \quad r < L,$$

$$u_r = \frac{QL^2}{2Hr}, \quad r > L.$$

Now we could find the azimuthal velocity, u_θ , from the momentum equation, but I find it conceptually cleaner to use the vorticity equation, which, linearized and steady, reduces to

$$f \nabla \cdot \mathbf{u} = -\alpha \zeta, \quad (3)$$

whose solution is

$$\zeta = \frac{-Qf}{H\alpha}.$$

Notice that this blows up as $\alpha \rightarrow 0$; there is in fact no steady solution in that case. We can find the azimuthal velocity from

$$\frac{1}{r} \frac{\partial}{\partial r} (ru_\theta) = \zeta,$$

which gives

$$u_\theta = \frac{-Qfr}{2H\alpha}, \quad r < L, \tag{4}$$

$$u_\theta = \frac{-QfL^2}{2H\alpha r}, \quad r > L. \tag{5}$$

In obtaining this solution we have required u_θ to be continuous at $r = L$; if we didn't, there would be infinite vorticity there, which would violate (3).

Now to find the perturbation height field, there are two ways. We can take the divergence of the momentum equation, which leads to a Poisson-type equation for h' in terms of the velocities, which are known. This can be solved if we have boundary conditions. This is the only way that works in general. However, because of the circular symmetry, we can do solve for h' directly from the radial momentum equation, which, linearized and steady, is

$$-fu_\theta = -g \frac{\partial h'}{\partial r} - \alpha u_r,$$

or

$$h' = \frac{1}{g} \int (fu_\theta - \alpha u_r) dr.$$

If we substitute in the solutions for $r < L$, we get

$$h' = \frac{-Qr^2}{4gH\alpha} (f^2 + \alpha^2) + C_1,$$

where C_1 is an integration constant. So far so good. Now going to $r > L$, we get

$$h' = \frac{-QL^2}{2H\alpha} (f^2 + \alpha^2) \ln r + C_2,$$

which blows up as $r \rightarrow \infty$! In fact there is no way around it, this solution is actually unacceptable. The problem is that we have assumed a steady solution, but we are continually removing mass from the domain, so a truly steady solution cannot exist. Even though the domain is infinite, supplying the mass from infinity requires a velocity which falls off as $1/r$, which is not fast enough to prevent the singular behavior. To make it work, we would really need to add a mass source somewhere which

would cancel the sink in the global average. This could be done, for example, by making the domain finite and adding a weak, spatially uniform mass source of the right magnitude to balanced the localized mass sink. The distributed source can be thought of as representing radiative cooling, while the stronger, localized sink represents convective heating (ask me in class and I'll ramble on more about this if you want).

However, in a sense this solution is not so bad; it is no worse behaved than, say, the point sources of potential flow theory, which have the same $1/r$ decay in the velocity field, and which would thus generate the same problem in the presence of either friction or rotation. Up to the point where we computed the height field, the physics is all qualitatively the same as it would be in the case where we balance mass properly. And, the solution for $r < L$ is valid regardless.

- (b) From the divergence equation (and the solution itself) we have the scaling for the radial velocity $u_r \sim QL/H$, while the azimuthal velocity scales as $u_\theta \sim QfL/2H\alpha$. The scaling for h' , using the solution for $r < L$, is $h' \sim QL^2f^2/gH\alpha$, if we assume $f \geq \alpha$. There are a number of different nonlinear terms we can use to estimate the importance of nonlinearity. In the mass conservation equation, we've neglected an advection term $u_r \partial h' / \partial r$. The ratio of that to Q , which we've retained, scales like

$$u_r \frac{\partial h'}{\partial r} \sim \frac{QL}{H} \frac{1}{L} \frac{QL^2f^2}{gH\alpha} = \left(\frac{Q^2L^2f^2}{gH^2\alpha} \right) Q^{-1} = \frac{QL^2f^2}{gH^2\alpha} = \frac{Q}{H\alpha} \frac{L^2}{L_d^2}.$$

So this suggests that nonlinearity will become important when the nondimensional quantity $\frac{Q}{H\alpha} \frac{L^2}{L_d^2}$ reaches order unity. In the radial momentum equation, we have a couple of nonlinear terms: an advection term, and a centrifugal term, u_θ^2/r . For $f > \alpha$, the azimuthal velocity exceeds the radial, so let's consider the latter one, and compare it to the Coriolis term that we retained, $-fu_\theta$. The ratio scales as

$$\frac{u_\theta^2/r}{fu_\theta} \sim \frac{u_\theta}{fL} \sim \frac{Q}{H\alpha}.$$

We see from the intermediate expression that this is basically a Rossby number. If $L \sim L_d$ this nondimensional number is the same as that obtained from the mass equation. For $L \gg L_d$, the momentum equation provides a stricter criterion for the onset of nonlinearity, vice versa for $L \ll L_d$.

- (c) If we combine the vorticity and mass conservation equations (including the mass source) to form a potential vorticity equation, the result is

$$\frac{d}{dt} \left(\frac{\zeta + f}{h} \right) = -\frac{Q}{h} \left(\frac{\zeta + f}{h} \right).$$

PV is no longer conserved in the presence of a mass source. Assuming steady state and circular symmetry — to make life easier — implies that the vorticity gradient is purely radial, which gives

$$\mathbf{u} \cdot \nabla P = u_r \frac{\partial P}{\partial r} = -\frac{QP}{h}, \quad (6)$$

where we've defined the potential vorticity $P = (\zeta + f)/h$ (sorry, I used Q for the mass source already... forgot to conserve notation). Now if we linearize the mass conservation equation we know, from our solution above, that $u_r > 0$ for $r > 0$, and vanishes at the origin. Thus P must be zero at the origin at least, since the advection term must vanish there. Apart from the origin, let's imagine $P > 0$. Then to satisfy (6) we must have $\partial P / \partial r < 0$, i.e., the origin is a local *maximum* of P . But that is inconsistent with our assumption of $P > 0$, since we already know $P = 0$ at the origin. The same argument holds if we assume $P < 0$. Thus the only possible solution is $P = 0$, which for h finite, implies $\zeta + f = 0$, or $\zeta = -f$. A state close to this is actually observed in the upper troposphere of both hurricanes and monsoons.

Here is another argument, which requires no linearization. The vorticity and height equations can be written in flux form:

$$\frac{\partial \zeta}{\partial t} + \nabla \cdot [\mathbf{u}(\zeta + f)] = 0, \quad (7)$$

$$\frac{\partial h}{\partial t} + \nabla \cdot (\mathbf{u}h) = Q. \quad (8)$$

Take steady state and integrate around a circular contour centered on the origin, and use Gauss' law to find

$$\oint [\mathbf{u}(\zeta + f)] \cdot \hat{\mathbf{n}} dl = 0, \quad (9)$$

$$\oint (\mathbf{u}h) \cdot \hat{\mathbf{n}} dl = \iint Q dA. \quad (10)$$

Since the vorticity contours are also circular contours centered on the origin, we can take the vorticity out of the integral in (9) to obtain

$$(\zeta + f) \oint \mathbf{u} \cdot \hat{\mathbf{n}} dl = 2\pi R u_r(R) (\zeta + f)|_R = 0.$$

where R is the radius of our contour, $u_r(R)$ is the radial velocity at $r = R$, and we have used the circular symmetry again to evaluate the integral. This result implies that one of two things must be true: either $\zeta + f = 0$, or $u_r(R) = 0$. From (10), we can see that the latter can't be true. An analysis similar to that done for the vorticity can be carried out, since the h contours are also circular, leading to

$$h \oint \mathbf{u} \cdot \hat{\mathbf{n}} dl = 2\pi R u_r(R) h|_R = \iint Q dA.$$

So as long as h itself does not vanish, for $Q \neq 0$, u_R cannot be zero, thus $\zeta + f$ must be.

While we have assumed circular symmetry in making this argument, it is not necessary to do so, it just makes things a little easier. Note also that we did not use the sign of Q . In fact the argument only works for $Q > 0$. If $Q < 0$, the radial velocity will be inward, and if there is a region of nonzero absolute vorticity outside the mass sink region, that vorticity will be advected in, and there is nothing to get rid of it. Even without considering advection, the zero absolute vorticity solution is *unstable* in the presence of a mass sink, *stable* in the presence of a mass source — this is easily shown but we're getting a bit far afield so we'll leave that out here, ask in class if interested.

The flow violates QG scaling because $\zeta = f$ implies $Ro = 1$. Also, depending on the value of the parameters, the divergence may be as large as the vorticity, which also implies large departure from geostrophic balance.

A.1 (Not from Vallis.) Plot on the same graph the shallow-water dispersion relations for quasi-geostrophic Rossby waves on a beta-plane and Poincare waves on an f plane. Use values for f and β valid at $45^\circ N$. Convince yourself that at any given spatial scale there is a large *scale separation in time* between the two waves, and thus that QG theory can be thought of as an asymptotic theory for the low-frequency component of the flow. (Ignore the fact that one involves β and the other doesn't, so that this comparison is not quite apples and apples.)

A.1 solution. Figure 1 shows a plot on a semilog scale (the two curves don't fit on one plot otherwise) for a fluid depth of 1 km (giving a deformation radius close to 1000 km, a reasonable value for the midlatitude atmosphere) and f_0 , β computed using earth parameters for a latitude of 45° . To simplify things, I set $l = 0$ in this plot.

5.4 Straightforward if tedious algebra, and we know what the result is.

5.11 (a) The shallow water momentum equation with linear drag is

$$\frac{d\mathbf{u}}{dt} + \mathbf{f} \times \mathbf{u} = -g\nabla\eta - \alpha\mathbf{u},$$

so when we take the curl to obtain the vorticity equation, we will have a term $-\alpha\zeta$ on the right hand side, and this will carry through to the QGPV equation. Thus the latter, linearized about a state of rest, is

$$\frac{\partial}{\partial t}(\nabla^2\psi - F\psi) + \beta\frac{\partial\psi}{\partial x} = -\alpha\nabla^2\psi. \quad (11)$$

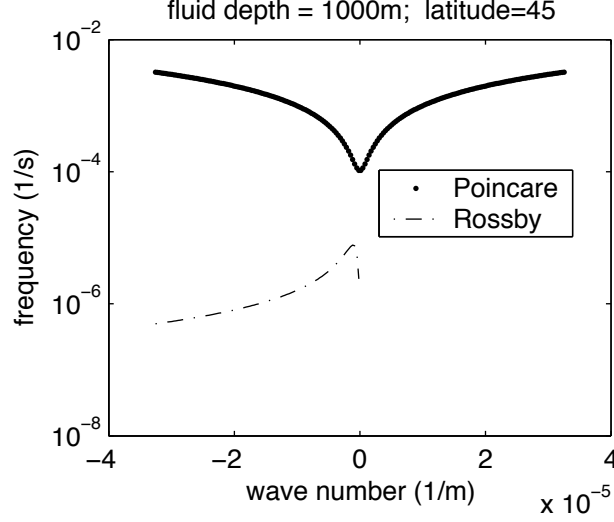


Figure 1: Dispersion relation for Poincare and Rossby waves, with $l = 0$.

Computing a dispersion relation gives

$$\omega = \frac{-k\beta - i\alpha K^2}{K^2 + F},$$

which differs from the inviscid case by the imaginary term in the numerator. Since the solution goes as $\exp(i\omega t)$, this term will always give *decay* in time, not growth, which makes physical sense. Note also that the damping rate goes to zero at $K = 0$ (this will be relevant in the next part).

- (b) Writing the 2-layer linear QGPV equations, and putting the drag term on the RHS of the equation for layer 2 only, assuming the usual wave like solutions, and doing some algebra, we get

$$\begin{aligned} \omega^2[K^4 + (F_1 + F_2)K^2] + \omega[\beta k(2K^2 + F_1 + F_2) + i\alpha K^2(K^2 + F_1)] \\ + k^2\beta^2 + i\alpha K^2 k\beta = 0. \end{aligned} \quad (12)$$

This can be solved analytically using the quadratic formula, but it gets a little messy. Figure 2 shows plots of the real and imaginary parts of ω for both modes, computed using matlab's "roots" function. This is done nondimensionally, so that f_0, F_1, F_2 are all 1, and to be consistent with our scaling then $\beta = 0.2$ (should be less than 1) and we use a "small" drag (really meaning small compared to f_0), $\alpha = 0.1$. Again I set $l = 0$ for simplicity here, so $k = K$. The frequencies look similar to their undamped counterparts, and the growth rates are negative, as we'd hope. The barotropic mode is damped more strongly at high wave numbers, most

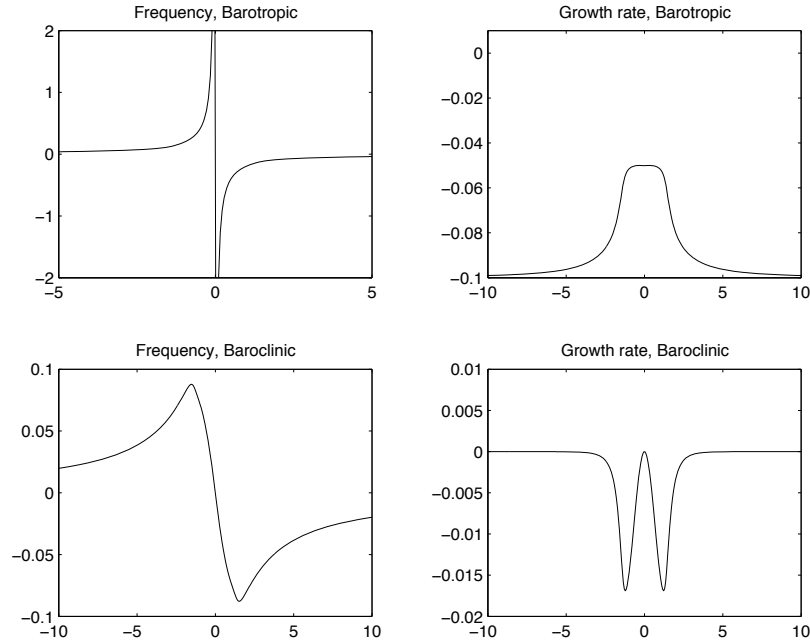


Figure 2: Dispersion relation for Rossby waves in a 2-layer system with linear drag in the bottom layer only. Units are dimensionless, so that the Coriolis parameter and deformation radii for both layers are equal to unity, $\beta = 0.2$, and $\alpha = 0.1$.

weakly at wave number zero. This makes some sense since the single-layer case, which we expect the barotropic mode to resemble, became undamped at $K = 0$. while the baroclinic mode is most strongly damped around $k = 1$, and is almost totally undamped at high wave numbers, a result for which I don't have a ready physical interpretation at this time, I'm still thinking about it. Mucho extra credit for he or she who comes up with one.